

Build Guide — Real Hardware Implementation

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Physical build of the 64× B200 inference cluster: site prep → rack & stack → power → cabling → switch bring-up → server bring-up → burn-in → handoff to software bootstrap.

Companion to [SPEC.md](#) (design), [docs/cabling.md](#) (cable tables), [docs/netbox.md](#) (SoT), [docs/bootstrap.md](#) (software bring-up — **starts where this guide ends**).

SoT rule: racks, devices, RU positions, IPs, and cables are authoritative in **NetBox** ([bootstrap/netbox/seed/site.yaml](#) pre-go-live). If this guide and NetBox disagree, fix NetBox, then re-export.

1 Scope

In scope	Out of scope
Site readiness (power/cooling/floor)	DC construction design → docs/facility.md
BOM summary & receiving	Full SKUs/pricing → docs/bom.md
Racking, power, cabling (169 cables)	RoCE/EVPN/QoS tuning → docs/network.md
Switch baseline config + ZTP	K8s / Metal3 / Flux bring-up → docs/bootstrap.md
Server firmware + rail/NIC verification	Model serving / storage tuning
Hardware burn-in + acceptance	Day-2 ops → docs/k8s.md

Baseline footprint: 4 racks — GPU-1, GPU-2, BOOT, STOR (SPEC §2A).

2 Safety

Hazard	Control
GPU server mass (8-GPU nodes are multi-person lifts)	Mechanical lift for anything > 40 kg; never solo-rack a B200 node
High-amperage 415 VAC A/B feeds	Licensed electrician for whips/PDUs; lockout-tagout during install
Laser (400G optics, AOC)	Dust caps on every unused optic; no eye-level inspection of live fibers
ESD	Wrist straps for all NIC/DIMM/GPU-adjacent work
Hot aisle (43–46 kW/rack)	No sustained work in contained hot aisle; hydrate, rotate staff
Airflow direction	Verify F-R vs R-F SKU per device before racking (leaves/spines/OOB ordered to match row)

3 Site readiness checklist

Complete **before** delivery day.

3.1 Power (SPEC §9)

Item	Requirement	Verify
Facility capacity	130–150 kW available (IT baseline ~100–115 kW + growth)	Utility/UPS schedule
GPU racks	High-density 415 VAC, A/B , ~50 kW/rack class	Per-rack whip rating
BOOT + STOR	Standard enterprise density, A/B	—
Redundancy	A/B to every rack; rPDUs dual-corded	A-side loss test planned
rPDUs	Metered/switched, C19/C13 mix per server PSU class	Ports $\geq$ device cords

3.2 Cooling (SPEC §10)

Tier	Requirement
Minimum	Hot-aisle containment + high-airflow CRAC sized for $2 \times \sim 46$ kW GPU racks
Preferred	Rear-door heat exchangers on GPU-1/GPU-2
Future	Layout must not block DLC retrofit (manifold path, CDU space)

3.3 Floor & logistics

- ☐ Floor loading rated for 4 fully populated racks (GPU racks are the heavy ones)
- ☐ Delivery path: dock → freight elevator → row, door/turn clearances for crated 8-GPU servers
- ☐ Staging area near row for unboxing + MAC/serial capture
- ☐ Cage/rack space: 4 adjacent rack positions, GPU racks in the high-kW zone (SPEC §2A.4)

3.4 Pre-staged data

- ☐ `bootstrap/netbox/seed/site.yaml` imported to design NetBox (or offline export run)
 - ☐ `docs/cabling.md` + `docs/cabling.csv` exported and printed
 - ☐ Cable labels printed **both ends** (`R{rail}-W{nn}`, `L{rail}S{spine}-U{n}`, `00B-{device}`)
 - ☐ Operator workstation: `kubectl`, `clusterctl`, `flux`, `yq`, `jq`, serial console, Redfish tooling
 - ☐ EOS images + Ubuntu 24.04 cloud image + Ironi IPA assets downloaded (see `docs/bootstrap.md` §5.1)
 - ☐ BMC credentials decided (env-injected later; **never** committed)
-

4 Bill of materials (summary)

Indicative quantities — exact SKUs in `docs/bom.md`.

Item	Qty	Notes
Dell PowerEdge XE9680 (HGX B200, 8× GPU)	8	8× 400G NIC (ConnectX-7 / BF3), 2× Xeon Scalable, 2 TB RAM, 8× 3.84 TB NVMe, 6U
Rail leaf — DCS-7060DX5-32	8	One per rail; match airflow to rack
Spine — DCS-7060DX5-64S	2	BOOT rack
OOB — DCS-7010TX-48	2	One per GPU rack
Mgmt server 1U (cp01–03)	3	32c / 256 GB / NVMe
Mgmt/utility 1U (seed01, util01–03)	4	seed may be repurposed util-class
Storage nodes / appliance	per <code>docs/storage.md</code>	Hot FS 150–300 TB + archive $\geq$ 750 TB usable
Racks 42–48U + rPDUs	4	GPU racks high-density rated
DAC QSFP-DD 400G 3m	64 + 20% spares	Host $\leftrightarrow$ leaf
AOC QSFP-DD 400G 15m	64 + 20% spares	Leaf $\leftrightarrow$ spine
Cat6 5m	14 + spares	BMC $\leftrightarrow$ OOB
Console server / KVM	1–2	BOOT rack
Optics spares	$\geq$ 20% of each type (SPEC §5.8)	

5 Phase 1 — Receive, stage, record

1. **Inspect on dock:** shock/tilt indicators on every crate; photograph damage before opening.
2. **Unbox in staging; capture identity into NetBox** for every device:
 - Serial number, asset tag
 - **BMC MAC** and **mgmt0 MAC** (workers: also note rail NIC card positions)
 - Update `mac_mgmt/BMC MACs` in `seed/site.yaml` → re-run `netbox-sync.sh --offline` (design seed is pre-go-live SoT; see `docs/netbox.md`)
3. **Firmware audit at staging** (faster than in-rack): record shipped BIOS/BMC/NIC/GPU firmware per node; flag deltas against baseline (§9.2).
4. **Label every cable both ends** from the printed sheets before anyone touches a rack.

Exit: NetBox devices = physical serials/MACs; labels printed; staging inventory signed off.

6 Phase 2 — Rack & stack

RU positions below match `seed/site.yaml` (NetBox is authoritative). General rules: heaviest low, rail leaves at top near host NICs (short DAC runs), OOB at rack top, patch/fiber management adjacent to leaves.

6.1 GPU-1 / GPU-2 (identical layout)

RU	Device	Notes
48	oob-sw1 (GPU-1) / oob-sw2 (GPU-2)	7010TX-48
42–39	leaf-rail0–3 (GPU-1) / leaf-rail4–7 (GPU-2)	7060DX5-32, rail index = NetBox <code>rail_index</code>
30, 24, 18, 12	worker01–04 (GPU-1) / worker05–08 (GPU-2)	XE9680, 6U each; mechanical lift
lower	rPDU A/B	Dual-cord every device

- Verify **airflow direction** matches row containment before bolting switches.
- Leave NIC-side clearance for 8× DAC per server; route via fiber managers, no kinks (DAC min bend radius).

6.2 BOOT

RU	Device
40	seed01
36–34	cp01–cp03
30–28	util01–util03
20, 18	spine1, spine2 (7060DX5-64S)
+	console server / KVM, patch panel

6.3 STOR

Per `docs/storage.md` — hot FS nodes low, archive nodes/shelves above, storage BMCs → oob-sw2.

Exit: all devices racked, torqued, airflow-checked; RU photos taken; NetBox rack elevations confirmed.

7 Phase 3 — Power

1. rPDUs on A/B feeds; **meter before load** (phase balance).
2. Power sequencing: **switches first, servers last** — OOB → spines/leaves → storage → cp/util → GPU workers (one rack at a time).
3. GPU rack sanity: power **one** worker, watch rPDU draw through POST (~1–2 kW), then enable the rest. Full rack $\approx$ 43–46 kW under load later; validate cooling with all 4 nodes at burn-in, not production.
4. Confirm PSU redundancy: pull A-side per rack (planned window) — no device drops.

Exit: all nodes/switches powered; rPDU meters green on both feeds; A/B failover proven.

8 Phase 4 — Cabling (169 cables)

Cable **in this order** — each class enables the next. Tables and labels: `docs/cabling.md` / `docs/cabling.csv`.

8.1 OOB + mgmt — 39× Cat6 + 2× DAC 25G (first!)

Per `cabling.md` §5, in order: **14× BMC** (VLAN 20), **15× server mgmt0** (VLAN 10, OS/PXE), **10× switch Ma1** (VLAN 20, ZTP), **2× OOB MLAG peer** (Ethernet49–50). OOB gives Redfish + ZTP reachability for everything after.

8.2 Leaf ↔ spine — 64× AOC 400G

Per leaf: Ethernet17–24 → 4× spine1 + 4× spine2, ports per `cabling.md` §4 ($L_{\text{rail}}S_{\text{spine}}-U_{\text{n}}$). Dress AOC with slack spools; these are 15 m runs to BOOT.

8.3 Host ↔ rail leaf — 64× DAC 400G

The rail identity is the whole design (SPEC §5.0): **GPU-*i* NIC of every worker → leaf-rail_*i* Ethernet{worker_index}** ($R_{\text{rail}}-W_{\text{nn}}$). Rail leaves sit in the same/near rack → 3 m DAC.

Rule	Check
GPU0/NIC0 → leaf-rail0 ... GPU7/NIC7 → leaf-rail7	Label audit per cable
worker0N → leaf EthernetN on all 8 leaves	Port map, not just link light
No host rail on the wrong leaf	Audit by label and LLDP (§10.4)

8.4 Post-cable audit

- ☐ All 169 labels scanned into NetBox; cable status → **connected**
- ☐ No DAC/AOC exceeding bend radius; no copper crossing power bundles
- ☐ Spares (20%) coiled, labeled, stored on-site

Exit: NetBox cable status **connected** = physical; photos per rack.

9 Phase 5 — Switch bring-up

All 12 switches provision via **Arista ZTP from seed01** — full design, rendered configs, and runbooks: docs/network.md. Nothing is hand-configured except seed01 and emergency console access.

9.1 OOB switches (oob-sw1/2) — zero-day bench ZTP

Per docs/network.md §3.2:

1. seed01 up on the bench (00-seed-host.sh done: dnsmasq + nginx + /var/www/html/ztp/).
 2. Serials of both 7010TXs into bootstrap/seed/ztp/serialmap.yaml → render.py → configs staged.
 3. Patch seed01 eth0 → oob-sw1 front port; power on → ZTP pulls config by serial → reloads.
 4. Repeat for oob-sw2. Rack both; cable access ports + MLAG peer (Ethernet49–50).
 5. Fallback: console server + paste the same rendered config (network.md §3.2 step 6).
- Dual uplink both 7010TXs = MLAG pair, VARP gateways 10.10.0.1 / 10.20.0.1; **no lossless QoS on OOB**.
 - Verify: show mlag active/live; show lldp neighbors sees every BMC (14) + mgmt0 (15) + Ma1 (10).

9.2 Spines + rail leaves — ZTP via OOB relay

Per docs/network.md §3.3: Ma1 → OOB VLAN 20 → relay → dnsmasq reservation (Ma1 MAC) → option 67 → ztp.py → per-serial config → reload. Spines first (convention), then all 8 leaves; BGP establishes as pairs complete.

1. Confirm reservations staged: grep ztp bootstrap/seed/generated/dhcp-hosts.conf = 10 lines.
2. Power spine1, spine2 → wait for show zerotouch = disabled → show bgp summary idle until leaves appear.
3. Power leaf-rail0–7 → ZTP; each leaf: 8 uplinks up/up, sessions Established to both spines.
4. Underlay + RoCEv2 AI profile arrive **in the rendered config** (PFC, ECN, watchdog, MTU 9216, eBGP) — do not improvise here; tuning deltas go through docs/network.md + render.py.

9.3 Switch verification gates

Gate	Command (EOS)	Pass
ZTP	show zerotouch	disabled on all 12; serial ↔ config match
OOB MLAG	show mlag / VARP	active/live; .1 answers both VLANs
Links	show interfaces status	8 down + 8 up per leaf connected
Optics	show interfaces transceiver	DOM in range, no flaps
Fabric	show bgp summary	All peers Established
Errors	show interfaces counters errors	0 CRC/FCS after 30 min
OOB	ping every BMC from oob-sw	14/14

Exit: 12/12 switches ZTP'd to target EOS, fabric up, zero-error soak 30 min.

10 Phase 6 — Server bring-up (via OOB/Redfish)

Do this **before** Metal3 touches anything — it derisks the CAPI run.

10.1 BMC baseline (all 15 servers)

- ☐ BMC reachable via OOB (<https://10.20.0.x> / Redfish)
- ☐ Set BMC admin credentials (from secrets, env-injected — matches BMC_USERNAME/BMC_PASSWORD plan)
- ☐ NTP = 10.10.0.10; syslog → ops; Redfish enabled; default passwords dead
- ☐ Boot mode UEFI, PXE first on **mgmt0** (10.10.x), virt/SR-IOV/ACS settings per GPU server vendor baseline

10.2 Firmware baseline

- ☐ BIOS + BMC at pinned versions (record in NetBox or build sheet)
- ☐ **NIC firmware** (ConnectX-7 / BF3) at target; RoCE enabled
- ☐ GPU firmware/VBIOS per NVIDIA baseline for the B200 platform
- ☐ NVMe firmware per vendor advisory

10.3 Per-worker hardware inventory (POST / BMC SEL)

- ☐ 8× B200 visible; NVLink fabric present
- ☐ 8× 400G NICs present, **PCIe locality matches GPU pairing** (NIC_i next to GPU_i)
- ☐ 2 TB RAM, 8× 3.84 TB NVMe, 2× boot NVMe RAID1
- ☐ BMC SEL clean (no correctable storms)

10.4 Rail verification (critical)

For each worker, each NIC_i must link to leaf-rail{i} Ethernet{worker_index}:

```
# per leaf – LLDP neighbor on EthernetN must be the right worker's right NIC
leaf-rail3# show lldp neighbors Ethernet4 detail # expect worker04 rail3 NIC
```

- ☐ 64/64 host links land on the labeled rail leaf+port (LLDP audit vs `cabling.csv`)
- ☐ All 64 links connected 400G full, FEC counters incrementing cleanly

Exit: every server Redfish-manageable, firmware pinned, 64 rails verified by LLDP.

11 Phase 7 — Burn-in & validation

Test	Tool	Target / pass
GPU health	<code>dcgmi diag -r 3</code> (post-OS, or vendor burn image)	No failures, all 64 GPUs
NVLink	<code>dcgmi nvlink / ven-</code> <code>dor tool</code>	All links up, no errors
GPU thermal/power	30–60 min GPU burn (all 8/node)	No throttle below spec; rack ΔT within cooling plan
Per-rail network	<code>perftest</code> (<code>ib_write_bw</code>) server-pair per rail	~400G line rate per rail, 0 retransmits of concern
Fabric soak	All rails loaded 2 h	ECN marks bounded, zero PFC storm / watchdog events
Storage attach	<code>fio</code> against hot FS from a worker	$\geq 10\text{--}20$ GB/s single-node (SPEC §6.4)
OOB soak	Continuous BMC polls 24 h	No flaps

Capture all results into the build record (attach to NetBox or ops wiki).

12 Phase 8 — Handoff to software bootstrap

Hardware is done when the [bootstrap acceptance criteria](#) can run. Hand off with:

Artifact	Where
NetBox = physical reality (serials, MACs, cables connected)	<code>netbox-sync.sh</code> export
Firmware/BIOS baseline record	Build sheet
Rail/LLDP audit output	Build sheet
BMC creds in secrets store (env path documented)	<code>docs/bootstrap.md</code> §5.1
Ubuntu/Ironic/EOS images staged on seed	<code>http://seed/images/</code>

Then run `docs/bootstrap.md` end-to-end: NetBox → seed → Metal3/CAPI (3 CP + 8 workers) → Flux platform → smoke. The interactive walkthrough of that flow is [demo/](#).

13 Acceptance checklist (hardware)

- ☐ 4 racks populated per NetBox elevations; airflow correct; rPDUs A/B metered
 - ☐ One-A-side power pull: zero device loss
 - ☐ 169/169 cables labeled, in NetBox **connected**, LLDP-verified (rails exact)
 - ☐ 12/12 switches target EOS; BGP Established; 30-min zero-error soak
 - ☐ 15/15 servers Redfish-reachable; firmware pinned; SEL clean
 - ☐ 64/64 GPUs pass `dcgmi diag -r 3`; 30-min full-rack thermal burn within spec
 - ☐ 8/8 rails ~400G perftest; 2-h fabric soak clean
 - ☐ Handoff artifacts complete; bootstrap §5.1 prerequisites all true
-

14 Troubleshooting

Symptom	Likely cause	Action
BMC unreachable	OOB cable/VLAN	Check <code>show lldp neighbors</code> on 7010TX; verify Cat6 label vs port
Rail link down	Wrong leaf/port or bad DAC	LLDP audit; swap DAC from spares; re-check label R{r}-W{n}
Leaf↔spine flapping	AOC/optic DOM out of range	<code>show interfaces transceiver</code> ; reseal/replace from spares
GPU missing in POST	Seating / power / firmware	Reseat node power; check SEL; re-flash GPU firmware baseline
NIC not 400G	Wrong cable type or port config	Confirm DAC 400G QSFP-DD; check interface speed/fec config
Rack thermal alarm in burn	Airflow mismatch or containment gap	Verify F-R/R-F SKUs; check blanking panels; call facilities
PFC pause storms in soak	QoS profile misapplied	Re-apply <code>docs/network.md</code> AI profile; verify ECN thresholds

15 References

Doc	Use
SPEC.md	Design authority (fabric, power, cooling, storage)
docs/cabling.md / cabling.csv	169-cable tables + labels
docs/netbox.md	SoT operations, day-0/day-2
docs/bootstrap.md	Software bring-up after this guide
docs/network.md	ZTP, underlay, RoCEv2 lossless profile (§9 gate)
docs/facility.md (planned)	Elevations, floor, cooling detail
demo/	Interactive walkthrough of the bring-up flow

16 Revision

Version	Date	Notes
0.1	2026-07-18	Initial hardware build guide